

Evolution of Sports Analytics

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HES 3000-002, Sport Performance Analytics

Why leverage data for decision making?

- Bias:
 - Memory & perception are unreliable
- Scale:
 - Humans can't track large samples
- Precision:
 - 96 mph $\neq$ “pretty fast”
- Complexity:
 - Too many interacting variables

Beginning of Sports Analytics



Analog Measurement (and Analysis)

Mid-1800s–1960s

- Sports began turning performance into **recorded information**.
- Data collection was manual, limited, and primarily descriptive.
- Scouting remained largely observational and qualitative.
- Limitations:
 - Box scores capture the *what* but entirely miss the how and why (i.e., context) – think back to opening exercise
 - High susceptibility to **confirmation bias** and **overgeneralization** from small samples

Analog Measurement (and Analysis)

BOSTON.						ATHLETIC.															
T.	R.	l.b.	P.O.	A.	E.	T.	R.	l.b.	P.O.	A.	E.										
G. Wright, s.s	6	4	4	1	5	2	Force, s. s...	5	1	2	1	3	2								
Leonard, 2b.	6	3	3	4	4	3	Eggler, c. f...	5	3	3	0	0	0								
O'Rourke, 1b	6	2	3	9	0	1	Fisler, r. f...	5	0	1	2	0	0								
Murnan, 1. f.	6	1	0	3	1	0	Meyerle, 3db	5	1	2	2	3	3								
Schafer, 2d b	6	3	3	3	1	2	Sutton, 1st b.	5	1	2	10	0	0								
McGinley, c.f	6	0	0	0	0	1	Coons, c....	5	1	0	1	1	3								
Manning, r.f.	6	0	2	2	0	0	Hall, l. f....	5	1	3	5	0	0								
Morrill, c....	6	2	2	4	1	2	Fowser, 2d b.	6	1	2	6	7	5								
Josephs, p..	5	4	4	1	1	2	Knight, p...	5	2	2	0	1	2								
Totals.... 53						19	21	27	13	13	Totals.... 46						11	17	27	15	15
Boston..... 9						1	3	3	4	1	0						2	5	19		
Athletic..... 1						0	0	0	3	3	2						2	0	11		
Runs earned—Boston, 4; Athletic, 5. Home-run—Hall, 1.																					
Total bases on hits—Boson, 22; Athletic, 20. First base by																					
errors—Boston, 8; Athletic, 5. Umpire, George White of																					
Lowell, Mass. Time 2h. 47m.																					

- **Henry Chadwick:**

- Helped standardize the baseball box score.

- **Harvey Pollack:**

- Expanded the statistics recorded in basketball.

- **Charles Reep:**

- Manually coded actions and passing sequences in soccer.
- Advocated for the long ball strategy

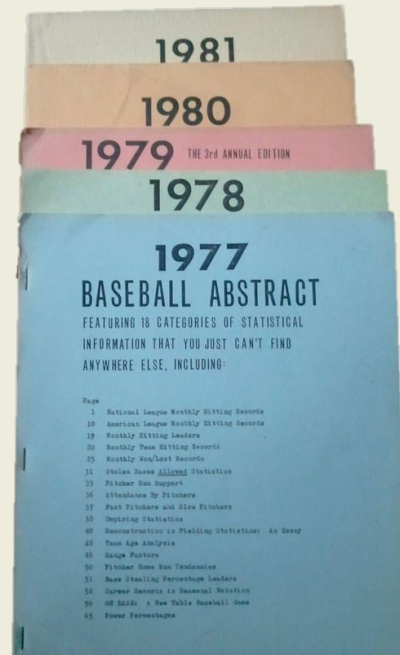
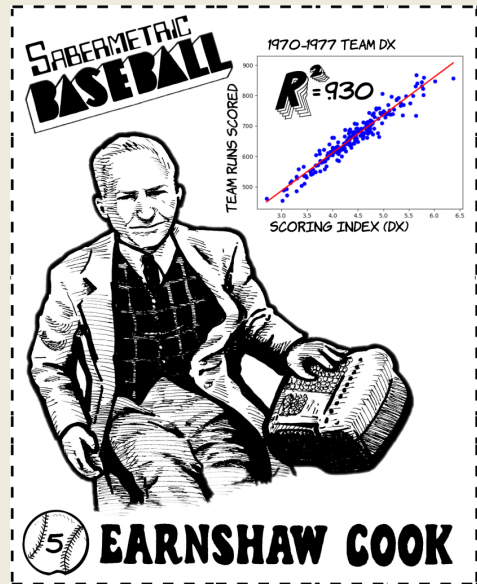


Challenging Conventional Wisdom

1960s–1990s

- Analysts began questioning accepted beliefs and whether traditional statistics actually measured performance.
- Greater emphasis on efficiency/rates, context, and contribution.
- Analysis began moving from recording outcomes to evaluating decisions.

Challenging Conventional Wisdom



- Earnshaw Cook:
 - Applied mathematical analysis to baseball strategy.
- Bill James:
 - Challenged traditional measures and popularized sabermetrics.
- Pete Palmer:
 - Developed methods for estimating runs and player value.

Why was analytics still limited?

Digital Measurement Revolution

1990s–2000s

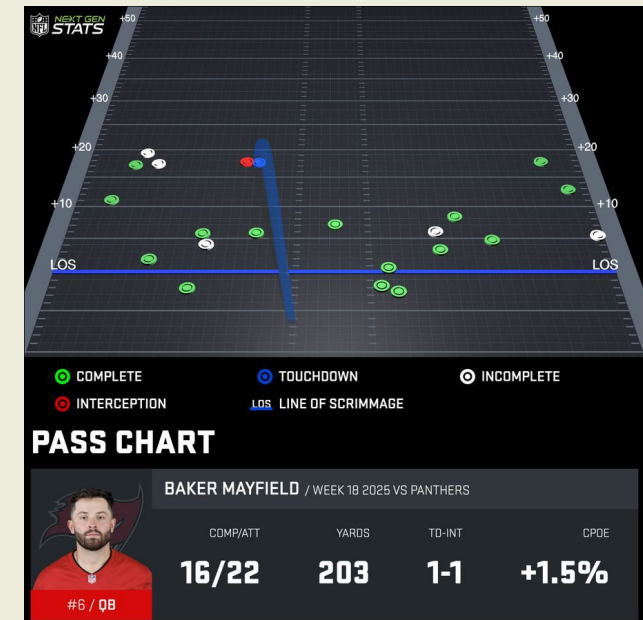
- New technologies dramatically expanded the volume, frequency, and detail of sports data.
 - Optical tracking, video analysis, and spatial tracking captured performance beyond traditional statistics.
 - Movement and physiological responses could increasingly be measured continuously.

Digital Measurement Revolution

- What are some examples of technologies that increased data in your sport?

Digital Measurement Revolution

- **Polar:** Portable heart-rate monitoring.
- **Garmin:** Wearable GPS live distance and pace numbers.
- **Catapult:** Spatial tracking systems for team sports
- **TrackMan:** Radar-based ball and object tracking.



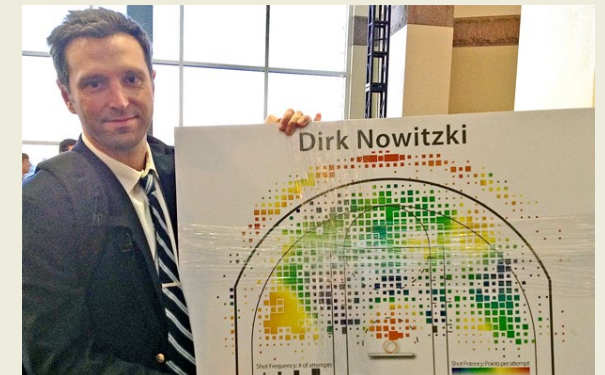
Modern Performance Analytics

2000s–Present

- Analytics moved from isolated outsiders toward front offices, coaching staffs, and dedicated departments.
- Analytics became integrated across scouting, strategy, athlete performance, and sports medicine.

Modern Performance Analytics

- Billy Beane and Paul DePodesta:
 - Applied sabermetric ideas to player acquisition and roster construction in Oakland.
- Daryl Morey
 - Built an analytically driven basketball operation around efficiency, probabilities, and decision-making.
- Kirk Goldsberry
 - Used spatial data and visualization to transform how basketball shooting and defensive performance were understood.



The Paradigm Shift:

- **The Traditional Mindset:**

- Heavy reliance on expert intuition, scouting lore, and decades of visual confirmation.
- The "gut feeling" approach to training and game planning.

- **Analytics Approach:**

- Uncovering hidden patterns that human cognitive biases naturally filter out or miss.
- Data doesn't replace the eye test; it **calibrates and enhances** it.

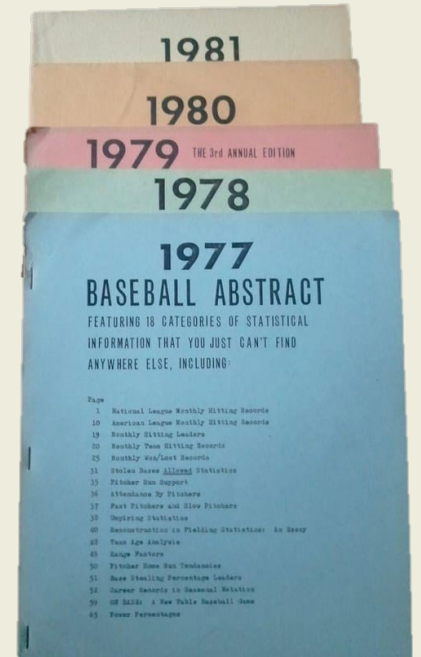
Evolution of Sports Analytics through *Moneyball*



Before Billie Bean and the A's

1. Challenging the Traditional Measures

- Batting average (BA) was traditionally treated as a central measure of offensive performance.
 - It captured hits but ignored other ways a player could reach base.
- **Bill James** questioned whether conventional statistics accurately represented player value.
 - Measures such as on-base percentage (OBP) helped shift attention toward the production of runs rather than traditional reputation.

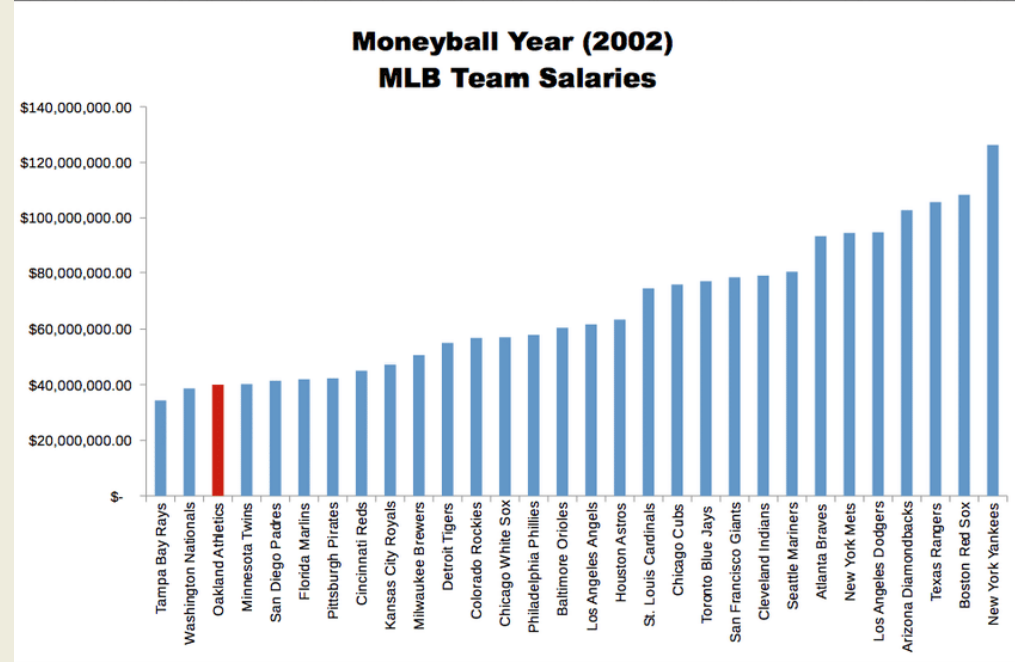


The first shift was not more data. It was questioning what the existing data actually measured.

Analysis to Implementation

2. The Oakland A's Put the Philosophy to Work

- The A's needed to build a competitive roster under substantial payroll constraints.
- Billy Beane, Paul DePodesta, and the Oakland front office looked for skills that contributed to winning but were undervalued by the player market.
 - Eg: OBP



What Happens When Everyone Catches Up?

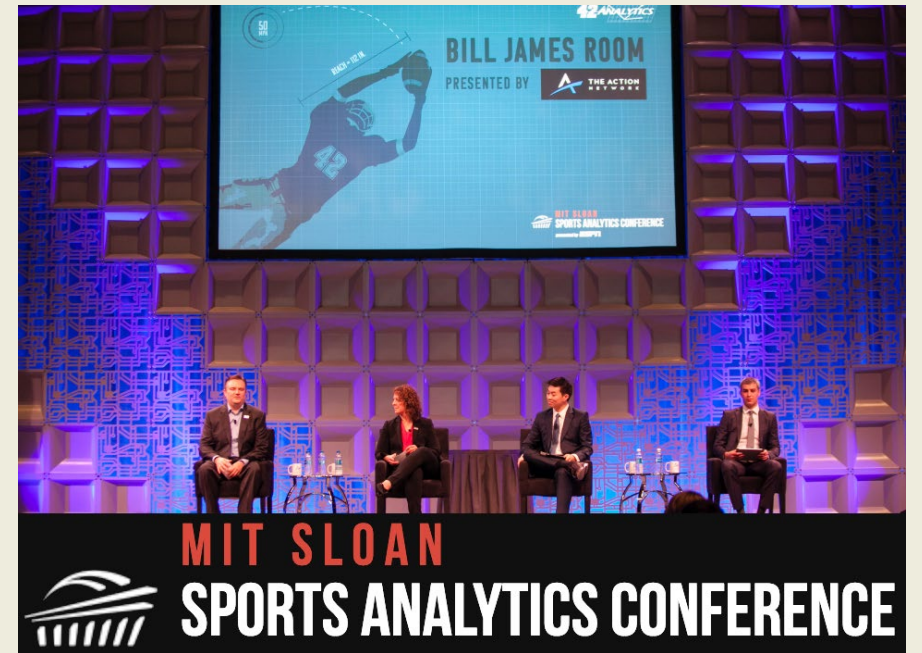
3. From Competitive Advantage to Widespread Adoption

- A's demonstrated analytical methods could reveal market inefficiencies.
- Other organizations expanded their own analytics capabilities.
- Previously undervalued skills became more widely recognized
 - (and the skill became appropriately priced)
 - **Competitive advantage** of OBP disappeared.

Beyond *Moneyball*

4. The Analytical Frontier Keeps Moving

- Modern analytics extends far beyond box scores and any single “advanced” statistic.
- Competitive advantage increasingly depends on asking better questions, **domain expertise**, and effective **communication / implementation**.



The Modern Sports Analytics Landscape

Performance analytics is now specialized into several “domains”:

- Scouting & Talent Evaluation
 - Objective profiling, tracking performance trajectories, and building predictive models for athlete recruitment.
- Athlete Tracking
 - Monitoring training and physical condition to manage training stress and prevent injuries.
- Strategy & Tactical Optimization
 - Leveraging situational probabilities and spatial efficiency to guide play-calling, defensive alignment, and game plans.

Summary & Looking Ahead

- **Key Takeaway:** Sport analytics continually evolves as new questions, data, technologies, and methods reshape how performance is understood and decisions are made.
- **Next Class:** Overview of the three domains (Tracking, Scouting, Strategy) and how data changes sports decisions.
- **For Next Time:** Think about a specific sports decision you've seen coaches make that you'd like to analyze with data.